

THEORY AND METHODS
WEIGHTING OF HOUSEHOLD SURVEY DATA:
A COMPARISON OF VARIOUS CALIBRATION,
INTEGRATED AND COSMETIC ESTIMATORS

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Summary: This paper compares the bias and efficiencies of different estimation techniques aimed at creating a unique set of case weights, which can be used to estimate characteristics for both person and household variables in a household sampling survey. The resulting estimates are compatible with external population information and thus attempt to correct for differential non-response (i.e. under-representation of certain parts of the population). The existing techniques of calibration and integrated weighting and cosmetic estimators are combined and modified to give improved weighted estimators. The theoretical properties of these estimators are outlined, and the performance of the estimators is investigated via simulation studies and by application to a data set. The effect of using auxiliary information on persons only, or on both persons and households, is evaluated. The performance of the estimators under condition of differential non-response has been simulated by using unequal probability sampling to mimic such conditions.

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1. Introduction

A household-based survey is generally conducted by selecting a sample of households and then including a single member, several or all members of the household, in the sample. In many studies estimates are required for both household and person characteristics, such as access to electricity and personal education levels. In obtaining these estimates and their standard errors, the sample design, coverage errors and non-response must be taken into account, as the estimates may otherwise be biased. This is usually achieved by assigning a weight to each sample unit, under certain assumptions concerning non-response.

At the first step, each sampled element is assigned a design weight, which is the inverse of the design inclusion probability. The second step of weighting usually attempts to compensate for non-response, by adjusting the design weights of the responding elements. Further weight adjustments are needed, as the sampled population often does not coincide with the target population, due to differential non-response (such as an over-representation of older females and under-representation of young males in the sample). This final step of weighting involves the use of auxiliary information, i.e. additional information on the study population such as that obtained from a census, to further improve the efficiency of estimates.

Two different sets of weights are generally provided for a survey, one for estimating the person-level characteristics, and one for estimating the household characteristics. Person weights are usually adjusted using post-stratification adjustment on the interlaced cells (cell weighting, i.e. the design-based approach) or calibration techniques (model-assisted approach), on available person level auxiliary information. This does not take into

consideration the fact that some persons belong to the same household. Household weights cannot be adjusted in a similar way, as reliable population information of household characteristics, other than geographical information, is generally not available. In practice, household weights are often derived from the person weights of the household members. Alexander (1987) adopts the principal person method (in which the household weight is based on the person designated as the 'principal person'), while Chowdhury (1994) uses the harmonic mean of the person weights to obtain the household weights. However, these methods of obtaining household weights still result in problems such as the totals (e.g. of household income) estimated by the two sets of weights, being inconsistent.

The integrated linear weighting procedure was proposed by Lemaître and Dufour (1987) and Nieuwenbroek (1993), to achieve one set of weights which would be appropriate for both person and household estimation, to overcome this problem of inconsistencies.

The present paper considers the case of a household survey using all household members. Section 2 considers the calibration technique, which is based on weighting at person level auxiliary information only. The integrated weighting technique is described in section 3, where the problem of defining person and household auxiliary variables at the level of persons is discussed. Extensions of the integrated weighting techniques to calibration and cosmetic estimators are discussed in sections 3.3 and 4. In order to evaluate the performance of the resulting estimators, while using person auxiliary variables only or person and household auxiliary variables simultaneously, a Monte Carlo simulation study is undertaken in section 5. Section 6 summarizes the findings of the study.

2. Calibration estimators

Consider a finite population U which consists of M households with a total of N persons, where a sample s of m households with a total number of n persons has been drawn with probability $p(s)$. Hence, if we suppose that sample household h has m_h household members, then $\sum_{h=1}^m m_h = n$. The inclusion probability of the k -th population element is $\pi_k = P(k \in s)$. In matrix notation, $\Pi = \text{diag}(\pi_k)$ is an $N \times N$ diagonal matrix of the inclusion probabilities. Let $d_k = 1/\pi_k$ denote the design weight for element k , compensated for unit non-response. The objective of many surveys is to estimate the population total $Y_T = \sum_{k \in U} y_k = \sum_U y_k$ where y_k is the value of the variable of interest, y , for the k -th element. Define an N -vector $\underline{Y} = (y_1, y_2, \dots, y_N)'$ of values of the study variable.

Auxiliary variables represent additional information about the study population. Assume the existence of J person-level auxiliary variables $x_1, x_2, \dots, x_j, \dots, x_J$ and define a J -vector $\underline{x}_k = (x_{k1}, x_{k2}, \dots, x_{kj}, \dots, x_{kJ})'$ for each $k \in U$, where x_{kj} denotes the value of the variable x_j for element k . The values of all the auxiliary variables are assembled into an $N \times J$ matrix $\mathbf{X}$ with $\underline{x}_k'$ in row k . The population totals for these variables, denoted by the J -vector $\underline{X}_T = \sum_U \underline{x}_k$, are assumed known. Vectors and matrices for the sample are denoted by subscript s . Thus, Π_s will be an $n \times n$ diagonal matrix, and $\mathbf{X}_s$ an $n \times J$ matrix with values of the auxiliary variables of the sample elements. Characteristics of persons are usually estimated using population counts for categorical variables such as sex, race, province etc., and the $n \times J$ matrix $\mathbf{X}_s$ contains dummy variables derived from these categorical auxiliary variables as illustrated in the following example.

Example 1:

Consider the first two households (h_1 and h_2) of a sample of m households, with four members ($h_1p_1, \dots, h_1p_4$) in household 1 and three members (h_2p_1, h_2p_2, h_2p_3) in the second household. The auxiliary variables AGE (divided into 4 groups: $A_1, \dots, A_4$) and SEX (M, F) are used. Suppose the values for the auxiliary information of the members specified in the $n \times J$ matrix $\mathbf{X}_s$,

$$\mathbf{X}_s = \begin{matrix} & \begin{matrix} A_1 & A_2 & A_3 & A_4 & M & F \end{matrix} \\ \begin{matrix} h_1p_1 \\ h_1p_2 \\ h_1p_3 \\ h_1p_4 \\ h_2p_1 \\ h_2p_2 \\ h_2p_3 \\ \dots \\ h_mp_{mh} \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ x_{n1} & x_{n2} & x_{n3} & x_{n4} & x_{n5} & x_{n6} \end{pmatrix} \end{matrix}$$

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Calibration estimation (Deville and Särndal, 1992, Deville et al., 1993) has become a widely used procedure for obtaining efficient estimates in sample surveys by using auxiliary information in the form of known population totals to produce a new set of weights. The calibration estimator is defined as $\hat{Y}_{cal} = \sum_s w_k y_k$ where the calibration weights w_k ($k \in s$) are as close as possible to the initial weights d_k , subject to a set of constraints $\sum_s w_k x_k = \underline{X}_T$. These constraints imply that the calibration weights will give perfect estimates when applied to each auxiliary survey variable.

Deville and Särndal (1992) consider a general distance function $G(w_k, d_k) = d_k v_k G(w_k/d_k)$, measuring the distance from the original weight d_k to the new weight w_k , with v_k a known positive weight unrelated to d_k . They describe the optimization problem as determining

new weights $w_k (k \in s)$ that minimize the average distance for the whole sample, i.e. $\text{Min}_{w_k} \sum_s G(w_k, d_k)$, subject to $\sum_s w_k \underline{x}_k = \underline{X}_T$. It follows that the calibration weights are $w_k = d_k F(\underline{x}'_k \underline{\lambda}_c / v_k)$, with $\underline{\lambda}_c = (\lambda_1, \dots, \lambda_j, \dots, \lambda_J)'$ a Lagrange multiplier vector determined by the above constraints, and F denoting the inverse function of $dG(\psi)/d\psi$, for $\psi = w_k/d_k$. The calibration estimator of Y is then $\hat{Y}_{cal} = \sum_s d_k F(\underline{x}'_k \underline{\lambda}_c / v_k) y_k$.

Under the generalized least square distance function with $G(w_k, d_k) = v_k d_k (w_k/d_k - 1)^2 / 2$ and $F(uv_k^{-1}) = 1 + uv_k^{-1}$, the calibrated weights are given by $w_k = d_k (1 + \underline{x}'_k \underline{\lambda}_c / v_k)$, and the calibration estimator becomes $\hat{Y}_{cal} = \hat{Y}_\pi + \sum_s d_k \underline{x}'_k \underline{\lambda}_c y_k / v_k$, where $\hat{Y}_\pi$ is the Horvitz-Thompson estimator. Deville and Särndal (1992) note that this is equivalent to the well-known generalized regression estimator (GREG), $\hat{Y}_r = \hat{Y}_\pi + (\underline{X}_T - \hat{\underline{X}}_\pi)' \hat{B}$ where $\hat{B} = (\mathbf{X}'_s \Pi_s^{-1} \mathbf{V}_s^{-1} \mathbf{X}_s^{-1} \mathbf{X}_s)^{-1} (\mathbf{X}'_s \Pi_s^{-1} \mathbf{V}_s^{-1} \underline{Y}_s)$. Here $\mathbf{V}_s = \text{diag}(v_k)$, with the v_k 's defined in relation to the variance structure of the specified model, ξ , defined as

$$y_k = \underline{x}'_k \beta + \varepsilon_k \quad \text{for } k \in U, \quad (1)$$

where $E_\xi(\varepsilon_k) = 0$, $\text{Var}_\xi(\varepsilon_k) = \sigma^2 v_k$ and $\text{Cov}_\xi(\varepsilon_k, \varepsilon_l) = 0$ for all $k \neq l$. The set of final weights for all the sample elements can be defined as $\underline{W}_s = \Pi_s^{-1} \underline{1}_n + \Pi_s^{-1} \mathbf{V}_s^{-1} \mathbf{X}_s (\mathbf{X}'_s \mathbf{V}_s^{-1} \Pi_s^{-1} \mathbf{X}_s)^{-1} (\underline{X}_T - \hat{\underline{X}}_\pi)$. This estimator thus falls into the model-assisted approach.

Deville and Särndal (1992) note that using

$G(w_k, d_k) = v_k d_k [w_k/d_k \log(w_k/d_k) - w_k/d_k + 1]$, $w_k/d_k > 0$ and $F(uv_k^{-1}) = \exp(u/v_k) > 0$ yields the exponential distance (also known as the multiplicative or raking ratio distance), where under this function the calibration weights are given by $w_k = d_k \exp(\underline{x}'_k \underline{\lambda}_c / v_k)$.

The literature contains a number of papers exploring various modifications of the technique, such as functional calibration (Estevao and Särndal, 2000),

where the w_k are now defined as a function of the design weights and two parameters, so that this class now includes all GREG estimators as special cases.

3. Integrated weighting technique

The shortcoming of using calibration techniques at person level is that the person weights so assigned will usually differ from person to person within the same household. This may cause uncertainty when household study variables are estimated, since no representative household weight exists. Furthermore, the estimators do not take household size into account, nor the fact that some persons belong to the same household, implying that the household should be treated as a cluster.

3.1 Integrated weighting based on person auxiliary variables only

The integrated linear weighting procedure has been developed to achieve one set of weights that overcomes these problems. Lemaître and Dufour (1987) propose the replacement of the person auxiliary variable values defined in the matrix $\mathbf{X}_s$ (where each person has a row vector with his/her own auxiliary information) in calibration, by a new $n \times J$ matrix $\mathbf{Z}_{pp}$. Here $\mathbf{Z}_{pp}$ is still based on person auxiliary variables (denoted by the subscript 'pp') but contains the averages for the corresponding auxiliary characteristics as entries. Since the auxiliary variables are generally categorical, and are thus coded into indicator variables, the $\mathbf{Z}_{pp}$ matrix will then contain the proportions for the household. Hence, the value for each member of household h , with a household size of m_h , is defined by $z_{hj} = \frac{a_{hj}}{m_h}$ where $a_{hj} = \sum_{k \in h} x_{kj}$ is the total for characteristic j in household h . Thus each person in the

household now has the same weight, but the weights for the household still add up to the same number as in the X matrix, for that household.

Example 1 continued:

The matrix $\mathbf{Z}_{pp}$ would be:

$$\mathbf{Z}_{pp} = \begin{matrix} & \begin{matrix} A_1 & A_2 & A_3 & A_4 & M & F \end{matrix} \\ \begin{matrix} h_1p_1 \\ h_1p_2 \\ h_1p_3 \\ h_1p_4 \\ h_2p_1 \\ h_2p_2 \\ h_2p_3 \\ \dots \\ h_mp_{m_h} \end{matrix} & \begin{pmatrix} \frac{1}{4} & \frac{2}{4} & 0 & \frac{1}{4} & \frac{3}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{2}{4} & 0 & \frac{1}{4} & \frac{3}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{2}{4} & 0 & \frac{1}{4} & \frac{3}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{2}{4} & 0 & \frac{1}{4} & \frac{3}{4} & \frac{1}{4} \\ 0 & \frac{1}{3} & \frac{2}{3} & 0 & \frac{1}{3} & \frac{2}{3} \\ 0 & \frac{1}{3} & \frac{2}{3} & 0 & \frac{1}{3} & \frac{2}{3} \\ 0 & \frac{1}{3} & \frac{2}{3} & 0 & \frac{1}{3} & \frac{2}{3} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ z_{m1} & z_{m2} & z_{m3} & z_{m4} & z_{m5} & z_{m6} \end{pmatrix} \end{matrix}$$

The interpretation of the weight $\frac{1}{4}$ for h_1p_1 is then that the person belongs to a household where $\frac{1}{4}$ of people are female, not that the person is $\frac{1}{4}$ female, $\frac{3}{4}$ male.



The question now is how the method of Lemaître and Dufour (1987) should be used when both person and household level auxiliary variables exist. Lemaître and Dufour (1987) only consider person level auxiliary variables such as sex and age group. Chowdhury (1994) discusses an application of inter alia the method of Lemaître and Dufour (1987). This model includes the variables sex and age group as well as household status (9 possible combinations of

adults and children), so the auxiliary variable matrix is presumably extended to contain indicator variables for these groups. Nieuwenbroek (1993) considers applying the approach of Lemaître and Dufour (1987) by modeling at the level of households, using person auxiliary variables only. It is not clear whether Chowdhury (1994) is modelling at person or household level.

3.2 Integrated weighting based on person and household auxiliary variables

The calibration technique, based on person auxiliary variables only, assigns different weights to members within the same household. Against this, the integrated weighting technique, which can be based on person-level auxiliary variables only or on both person and household auxiliary variables simultaneously, allocates the same final weight to all members in a household. The question to be addressed is whether the combination of these techniques, applied at person level, is useful for estimation at person level and household level totals. We extend the previous matrices $\mathbf{Z}_s$ by adding columns for each category of the household auxiliary variables under consideration. For example 1, suppose we add the variable ‘Area’, divided into categories Urban_formal, Urban_informal and Rural. We define three new variables – U_f, U_i, R . The entries of the new area variables would be the inverse of the household size for the area variable to which the household belongs, and 0 for the other two area variables. This $\mathbf{Z}$ matrix that includes both person and household auxiliary variables will be denoted as $\mathbf{Z}_{ph}$.

Example 1 continued:

The matrix $\mathbf{Z}_{ph}$ would be:

$$\mathbf{Z}_{ph} = \begin{matrix} & \begin{matrix} A_1 & A_2 & A_3 & A_4 & M & F & U_f & U_i & R \end{matrix} \\ \begin{matrix} h_1p_1 \\ h_1p_2 \\ h_1p_3 \\ h_1p_4 \\ h_2p_1 \\ h_2p_2 \\ h_2p_3 \\ \dots \\ h_mp_{m_h} \end{matrix} & \begin{pmatrix} \frac{1}{4} & \frac{2}{4} & 0 & \frac{1}{4} & \frac{3}{4} & \frac{1}{4} & \frac{1}{4} & 0 & 0 \\ \frac{1}{4} & \frac{2}{4} & 0 & \frac{1}{4} & \frac{3}{4} & \frac{1}{4} & \frac{1}{4} & 0 & 0 \\ \frac{1}{4} & \frac{2}{4} & 0 & \frac{1}{4} & \frac{3}{4} & \frac{1}{4} & \frac{1}{4} & 0 & 0 \\ \frac{1}{4} & \frac{2}{4} & 0 & \frac{1}{4} & \frac{3}{4} & \frac{1}{4} & \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{3} & \frac{2}{3} & 0 & \frac{1}{3} & \frac{2}{3} & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{3} & \frac{2}{3} & 0 & \frac{1}{3} & \frac{2}{3} & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{3} & \frac{2}{3} & 0 & \frac{1}{3} & \frac{2}{3} & 0 & 0 & \frac{1}{3} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ z_{m1} & z_{m2} & z_{m3} & z_{m4} & z_{m5} & z_{m6} & z_{m7} & z_{m8} & z_{m9} \end{pmatrix} \end{matrix}$$

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3.3 Integration through calibration

In order to obtain the weights, we proceed similarly to Deville and Särndal (1992). Consider the general distance function, $G(w_k, d_k)$, and the set of constraints $\sum_{k \in s} w_k z_k = \underline{X}_T$, where $z_k = (z_{h1}, z_{h2}, \dots, z_{hj}, \dots, z_{hJ})'$ now relates to the auxiliary variables for person k belonging to household h . Minimizing $\sum_{k \in s} G(w_k, d_k) - \lambda'_c (\sum_{k \in s} w_k z_k - \sum_{k \in U} z_k)$ with respect to the w_k . (where $\sum_{k \in U} z_k = \sum_{k \in U} x_k = \underline{X}_T$) leads to the calibration-integrated weights $w_k = d_k F(z'_k \lambda_c / v_k)$ and the estimator of Y :

$$\hat{Y}_z = \sum_{k \in s} w_k y_k = \sum_{k \in s} d_k F(z'_k \lambda_c / v_k) y_k.$$

For the linear distance function, the calibration-integrated weights based on the $\mathbf{Z}_s$ matrix ($\mathbf{Z}_{pp}$ or $\mathbf{Z}_{ph}$) are given by $w_k = d_k (1 + z'_k \lambda_c / v_k)$, where the Lagrange multiplier λ_c is determined as the solution of the system

$(\sum_{k \in s} d_k z_k z'_k / v_k) \lambda_c = \underline{X}_T - \hat{\underline{X}}_\pi$. This yields

$$\hat{Y}_{zcal} = \hat{Y}_\pi + (\underline{X}_T - \hat{\underline{X}}_\pi)' \hat{B}_z = \hat{Y}_T,$$

the GREG estimator based on the $\mathbf{Z}_s$ matrix being

$$\hat{B}_z = (\mathbf{Z}'_s \Pi_s^{-1} \mathbf{V}_s^{-1} \mathbf{Z}_s)^{-1} (\mathbf{Z}'_s \Pi_s^{-1} \mathbf{V}_s^{-1} \underline{Y}_s). \quad (2)$$

Note that in this case the GREG estimator is defined under the model ξ , defined in (1) with $\underline{x}_k$ replaced by $\underline{z}_k$. The $n \times 1$ person level vector of weights is

$$\underline{W}_s = \Pi_s^{-1} \underline{1}_n + \Pi_s^{-1} \mathbf{Z}_s (\mathbf{Z}'_s \Pi_s^{-1} \mathbf{Z}_s)^{-1} (\underline{X}_T - \hat{\underline{X}}_\pi)$$

where $\underline{1}_n$ is an n -vector with ones as entries. Since each household member obtains the same row vector of auxiliary information in matrix $\mathbf{Z}_s$, each person in a household will have the same final weight if the household design weight is allocated to all household members. Thus, this weight will also be the appropriate weight to be used for household study variables.

4. Cosmetic and cosmetic-integrated estimators

The cosmetic estimator, expressed by Brewer (1999) in a design-based as well as a model-based form, is defined as

$$\begin{aligned} \hat{Y}_{\text{cos}} &= \hat{Y}_\pi + (\underline{X}_T - \hat{\underline{X}}_\pi)' \hat{B}_{\text{cos}} \\ &= \underline{Y}_s + (\underline{X}_T - \underline{X}_{Ts})' \hat{B}_{\text{cos}} \end{aligned} \quad (3)$$

where $\underline{X}_{Ts} = \mathbf{X}'_s \underline{1}_n$, a $J \times 1$ vector with sample totals of the auxiliary variables,

$$\hat{B}_{\text{cos}} = [\mathbf{X}'_s \mathbf{C}_s^{-1} (\Pi_s^{-1} - I_n) \mathbf{X}_s]^{-1} [\mathbf{X}'_s \mathbf{C}_s^{-1} (\Pi_s^{-1} - I_n) \underline{Y}_s]$$

and $\mathbf{C}_s \underline{1}_n = \mathbf{X}_s \alpha$ any linear combination of the columns of $\mathbf{X}_s$.

The aim of these estimators is to obtain the best properties of the design-based approach as well as the model-based approach, particularly as far as variance estimation is concerned.

The intersection of the cosmetic estimator with the calibration estimator was derived by Brewer (1999). This so-called cosmetic calibration (COSCAL) estimator can be seen as an intersection of the cosmetic estimators of Särndal and Wright (1984) with the calibration estimators of Deville and Särndal (1992) and is defined as

$$\hat{Y}_{\text{cos cal}} = \underline{W}'_{cc} \underline{Y}_s,$$

with $\underline{W}_{cc}$ equal to

$$\underline{W}_{cc1} = \Pi_s^{-1} \underline{1}_n + (\Pi_s^{-1} - \mathbf{I}_n) \mathbf{C}_s^{-1} \mathbf{X}_s [\mathbf{X}'_s (\Pi_s^{-1} - \mathbf{I}_n) \mathbf{C}_s^{-1} \mathbf{X}_s]^{-1} (\underline{X}_T - \hat{\underline{X}}_\pi)$$

or

$$\underline{W}_{cc2} = \underline{1}_n + (\Pi_s^{-1} - \mathbf{I}_n) \mathbf{C}_s^{-1} \mathbf{X}_s [\mathbf{X}'_s (\Pi_s^{-1} - \mathbf{I}_n) \mathbf{C}_s^{-1} \mathbf{X}_s]^{-1} (\underline{X}_T - \underline{X}_{Ts}),$$

according to the two equivalent expressions for the estimator in (3). This estimator can be extended to integrated weighting to obtain a cosmetic calibration-integrated estimator (ZCOSCAL), by replacing the factor $\Pi_s^{-1} \mathbf{V}_s^{-1}$ in the ZCAL estimator (2) by factor $\mathbf{C}_s^{-1} (\Pi_s^{-1} - \mathbf{I}_n)$ to obtain

$$\begin{aligned} \hat{Y}_{z \text{ cos}} &= \underline{1}'_n \Pi_s^{-1} \underline{Y}_s + (\underline{1}'_N \mathbf{Z} - \underline{1}'_n \Pi_s^{-1} \mathbf{Z}_s) \hat{B}_{z \text{ cos}} \\ &= \underline{1}'_n \underline{Y}_s + (\underline{1}'_N \mathbf{Z} - \underline{1}'_n \mathbf{Z}_s) \hat{B}_{z \text{ cos}} \end{aligned} \quad (4)$$

where

$$\hat{B}_{z \text{ cos}} = [\mathbf{Z}'_s \mathbf{C}_s^{-1} (\Pi_s^{-1} - \mathbf{I}_n) \mathbf{Z}_s]^{-1} [\mathbf{Z}'_s \mathbf{C}_s^{-1} (\Pi_s^{-1} - \mathbf{I}_n) \underline{Y}_s]. \quad (5)$$

The proof follows by replacing $\mathbf{X}_s$ with $\mathbf{Z}_s$ and noting that expressions in (4) are equal when $\underline{1}'_n (\Pi_s^{-1} - \mathbf{I}_n) (\underline{Y}_s - \mathbf{Z}_s \hat{B}_{z \text{ cos}}) = 0$ (Neethling, 2004).

Since $\underline{1}'_N \mathbf{Z} = \underline{X}'_T$, $\underline{1}_n \Pi_s^{-1} \mathbf{Z}_s = \hat{\underline{X}}'_\pi$ and $\underline{1}'_n \mathbf{Z}_s = \underline{1}'_n \mathbf{X}_s = \underline{X}'_{Ts}$, the equations in (4) can be rewritten as

$$\begin{aligned}\hat{Y}_{z \cos} &= \hat{Y}_\pi + \left(\underline{X}_T - \hat{\underline{X}}_\pi \right)' \hat{B}_{z \cos} \\ &= Y_s + (\underline{X}_T - \underline{X}_{Ts})' \hat{B}_{z \cos}\end{aligned}$$

respectively, with $Y_s = \underline{1}'_n \underline{Y}_s$ and $\hat{B}_{z \cos}$ defined in (5).

The ZCOSCAL weights for the design-based estimator can be rewritten as

$$w_{k_{zcc1}} = d_k (1 + \underline{z}'_k \underline{\lambda}_c / v_k),$$

where $\underline{\lambda}_c = (\sum_s d_k \underline{z}_k \underline{z}'_k / v_k)^{-1} (\underline{X}_T - \hat{\underline{X}}_\pi)$ and $v_k^{-1} = (1 - \pi_k) / c_k$. For the model-based approach,

$$w_{k_{zcc2}} = (1 + \underline{z}'_k \underline{\lambda}_c^* / v_k),$$

with $\underline{\lambda}_c^* = (\sum_s \underline{z}_k \underline{z}'_k / v_k)^{-1} (\underline{X}_T - \hat{\underline{X}}_\pi)$ and $v_k^{-1} = (d_k - 1) / c_k$.

The ZCOSCAL weights can be easily calculated using the software package CALMAR, developed by the French national statistics agency (I.N.S.E.E). Cosmetic estimators combine both design-based and model-assisted estimators. Both these options will yield the same result when, for the cosmetic design-based estimators, the supplementary weighting variable are defined as

$$v_k^{-1} = (1 - \pi_k) / c_k$$

and the initial weight as d_k . For the model-based estimators the initial weights must be set equal to 1, while the supplementary weighting variables becomes

$$v_k^{-1} = (d_k - 1) / c_k.$$

In household surveys, all households have inclusion probabilities that are small compared with one. Since the difference between matrices Π_s^{-1} and $(\Pi_s^{-1} - I_n)$ tends to be trivial, it would be expected that the cosmetic estimators will vary only trivially from the ordinary CAL and ZCAL estimators. However, the reasons for using cosmetically calibrated or integrated estimators lie in its simultaneous support by design-based and

model-based inference and in the simplification of the estimators of relevant variances (model-based, design-based and anticipated) (Brewer, 1999 and Neethling, 2004).

5. Simulation study

The Monte Carlo simulation study used to compare the performance of the estimators was based on the 10% sample of the 1996 census, compiled by Statistics South Africa (Stats SA). The sample for one of South Africa's provinces, Gauteng, was considered as the population. Persons from special institutions and hostels as well as homeless persons were excluded for the purpose of the study. The final 'population' is 176 386 households with 602 647 persons.

For the simulation study we used 200 household samples of size 4000 each. In order to simulate the type of non-response found in practical situations (and since we are looking at the case where all persons in the selected households are included in the sample) the non-response structure was brought in at household level. Furthermore, the non-response had to apply at the person level as well, as we are comparing the different techniques for person level study variables as well as for household level study variables. Therefore, for each sample the households were drawn with different selection probabilities, based on age and race of the 'Head of the household'. The explanation of the simulated non-response is given in Table 1 below. The abbreviation 'Pop' refers to the population figures while 'Sp' presents the sample composition. For example, the cell: (Black, ≥ 60 years) indicates that although 7.53% of all the heads of households in the population fell in this cell, a larger proportion, 11.0%, of the heads of households in the sample belonged to this cell.

Table 1. Information for the head of the household for the sample and population

Age Group	Race			Total
	Black	Coloured/ Asian	White	
≤ 24 years	Pop: 4.32% (Sp: 5.5%)	Pop: 0.28% (Sp: 1.0%)	Pop: 1.46% (Sp: 3.5%)	Pop: 6.07% (Sp: 10.0%)
25–44 years	Pop: 38.82% (Sp: 22.0%)	Pop: 2.96% (Sp: 4.0%)	Pop: 13.22% (Sp: 14.0%)	Pop: 55.0% (Sp: 40.0%)
45–59 years	Pop: 15.14% (Sp: 16.5%)	Pop: 1.35% (Sp: 3.0%)	Pop: 8.2% (Sp: 10.5%)	Pop: 24.7% (Sp: 30.0%)
≥ 60 years	Pop: 7.53% (Sp: 11.0%)	Pop: 1.0% (Sp: 2.0%)	Pop: 6.09% (Sp: 7.0%)	Pop: 14.24% (Sp: 20.0%)
Total	Pop: 65.81% (Sp: 55.0%)	Pop: 5.21% (Sp: 10.0%)	Pop: 28.97% (Sp: 35.0%)	
Average hh size	3.9	4.2	2.8	3.8

The reason for assessing a sample with nonresponse errors is twofold. Firstly, evaluating the results on a random sample may result in the comparison of the performance of the estimators being more an evaluation of the quality of drawing the sample than a comparison of the estimators. Secondly, this will ensure that the weighting procedures are evaluated under non-perfect circumstances. Since a researcher will usually not be aware of all nonresponse errors, the samples were considered as if no nonresponse errors existed. The question then is which technique and which set of auxiliary information would be the best to use under practical circumstances.

Two sets of auxiliary variables were used in the simulation study:

- Person level auxiliary variables only (indicated by “pp”):
 - ▶ AGE (divided into 6 categories: 0–14, 15–24, 25–34, 35–59, 60+)
 - ▶ SEX (2 categories)
 - ▶ RACE (4 categories)
- Person level as well as household level auxiliary variables (indicated by “ph”):

- ▶ The three person level auxiliary variables
- ▶ AREA: (3 categories: Urban_formal, Urban_informal, Rural)

The population totals of the following study variables were estimated:

- Three person level study variables:
 - ▶ UNEMPL => (Number of people stating that they were unemployed)
 - ▶ SIGHT => (Number of persons with a sight disability)
 - ▶ YEARS5 => (Number of persons aged 5 years and younger)
- Three household level study variables:
 - ▶ DWELGR => (Number of Houses / brick structures on separate stand)
 - ▶ COOK => (Number of households using electricity for cooking)
 - ▶ WATER => (Number of households with piped water in dwelling or on site)

The basic assumption is that the chosen auxiliary variables will be useful in improving the estimation from the given sample, if they are correlated with the study variable(s) and thus carry information about the study variable(s). This was evaluated here using Cramer's V (sometimes called Cramer's Phi), a contingency table correlation coefficient, as a measure of association, calculated from the population data. Howell (2002) indicates that this coefficient is applicable to tables larger than 2×2 and that the phi coefficient for 2×2 tables is a special case of the Cramer's coefficient with the sign preserved. This gives the following results:

Table 2. Cramer's Phi values between auxiliary variables and study variables

	Unempl	Sight	Years5
Age groups	0.5860	0.0845	0.5895
Sex	0.1453	0.0252	0.0054
Race	0.0857	0.0494	0.0546

	Dwelgr	Cook	Water
Area	0.6522	0.4769	0.4207

Much of the literature deals with a single study variable, whereas in practice there are usually several variables of interest. The practical problem arises in obtaining a set of auxiliary variables that is correlated to most of the study variables.

The estimated total, $\hat{Y}$, and the estimated variance of $\hat{Y}$ were computed for each of the 200 samples. The different techniques were compared using different sets of weights and auxiliary variables, as summarized in Table 3. The abbreviations of the different techniques are defined in the column 'Name'. Techniques based on the generalized least square distance function are indicated by 'linear method' while those based on the exponential distance function (also called the raking ratio distance), are indicated by 'raking ratio method' as described in section 2.

Table 3. Abbreviations used for the estimation techniques

Name	Technique	Auxiliary variables	Final weight, w_k
HWDISTR	Direct weighting up to number of households per district	None	Design weight
CAL_pp	Calibration, linear method	Person	Calibration (linear)
CALr_pp	Calibration, raking ratio method	Person	Calibration (raking ratio)
ZCAL_pp	ZCAL, linear method	Person	Calibration (linear)
ZCALr_pp	ZCAL, raking ratio method	Person	Calibration (raking ratio)
ZCOS_pp	ZCOSCAL	Person	Cosmetic
ZCAL_ph	ZCAL, linear method	Person and household	Calibration (linear)
ZCALr_ph	ZCAL, raking ratio method	Person and household	Calibration (raking ratio)
ZCOS_ph	ZCOSCAL	Person and household	Cosmetic

Several summary measures are examined for the estimators of the totals.

- Expected value of $\hat{Y}$ (i.e. the average estimated total):

$$Av(y) = \bar{\hat{Y}} = E(\hat{Y}) = \frac{1}{R} \sum_{r=1}^R \hat{Y}_r$$

where R indicates the number of replications and $\hat{Y}_r$ the estimated total obtained for r -th replicated sample.

- The average bias of the estimator:

$$AvBias = \frac{1}{R} \sum_{r=1}^R (\hat{Y}_r - Y)$$

where Y is the true population total.

- Descriptive statistics of the bias: $\hat{Y} - Y$ (average, minimum, maximum, standard error), as well as a test for normality (Shapiro-Wilk test), included to check for skewness of the bias distribution.
- The percentage relative bias of the estimator, with respect to the population value:

$$\% Rel Bias = \frac{E(\hat{Y}) - Y}{Y} \times 100$$

In order to allocate different but still meaningful design weights for persons, the number of households per magisterial district was used to define the household design weight for a household in district i , $HWDISTR$, is defined as

$$d_{h_i} = \frac{\text{Number of households in district } i, \text{ in the population}}{\text{Number of households in district } i, \text{ in the sample}}.$$

When using these weights as the final weights of the household members, without any adjustments, a problem occurs in that the estimated total of persons deviates from the population size of 602647 (e.g. 632665.89 for the first sample in the simulation study). To obtain comparable results for person study variables, these weights are adjusted by the necessary factor (e.g. $602647/632665.89 = 0.95255$) when used as final weights.

The results of the person level study variables are considered in section 5.1, and estimators for household study variables in section 5.2.

5.1 Simulation results of estimators for person level study variables

The information given in Table 4, which shows the descriptive statistics and the Shapiro-Wilk test for normality (prob: normal) of the bias of these variables for the different estimation methods, can be summarized as follows:

- The integrated weighting estimators based on ‘pp’ auxiliary variables yield approximately the same results. The same conclusion can be made for the ‘ph’ auxiliary variable cases as well as for the calibration techniques.
- The bias of UNEMPL and YEARS5 are the smallest for integrated weighted techniques with ‘ph’ auxiliary variables.
- Although the average bias of UNEMPL and SIGHT are positive, the minimum values are negative, which indicate that underestimation appears for some samples but not on average.
- Most of the distributions of the bias of the estimates do not differ significantly from the normal distribution (see Column: ‘prob: normal’). Only the distributions of the integrated weighting estimators for SIGHT deviate slightly from normality with p-values of approximately 0.04.
- It was found that the average bias values of all the estimates differ significantly from zero (Neethling, 2004).

The standard error values obtained for the estimated population totals are given in Table 5, for the case of unequal design weights (HWDISTR).

Table 4. Simulation measures obtained for person level variables:
Descriptive statistics of BIAS = (estimated total – population total)

Method	UNEMPL				YEARS				SIGHT			
	Average (AvBias)	Max	Min	Prob: normal	Average Av(Bias)	Max	Min	Prob: normal	Average (AvBias)	Max	Min	Prob: normal
HWDISTR	11846	18190	3489	0.8272	-2260	1857	-6233	0.8280	666	3093	-2821	0.1211
CAL_pp	3483	9719	-2162	0.9784	-1608	1837	-6084	0.5685	528	3057	-217	0.0873
CALr_pp	3671	9802	-1949	0.9693	-1591	1891	-6073	0.5664	458	3007	-294	0.0990
COS_pp	3486	9723	-2154	0.9779	-1608	1838	-6085	0.5679	527	3058	-218	0.0877
ZCAL_pp	2962	9171	-2897	0.8470	-1169	2178	-5600	0.4836	829	3400	-2274	0.0430
ZCALr_pp	2566	8631	-3242	0.7500	-1052	2295	-5512	0.4520	775	3344	-2253	0.0399
ZCOS_pp	2964	9173	-2895	0.8483	-1169	2179	-5600	0.4832	828	3401	-2274	0.0430
ZCAL_ph	508	7011	-4885	0.9633	-953	2485	-5493	0.5080	998	3376	-2023	0.0433
ZCALr_ph	583	7105	-4861	0.9823	-906	2440	-5437	0.4773	899	3308	-2033	0.0443
ZCOS_ph	509	7008	-4880	0.9626	-953	2485	-5493	0.5068	997	3377	-2023	0.0436

Table 5. Standard errors of simulation study estimators for the totals of person study variables

Method	Std. Error		
	Unempl	Years5	Sight
HWDISTR	185.2140	108.6923	82.7443
CAL_pp	154.8037	89.4310	85.1421
CALr_pp	154.4778	89.5975	84.9689
COS_pp	154.8069	89.4320	85.1393
ZCAL_pp	158.3243	89.6768	84.6841
ZCALr_pp	157.8913	90.0901	84.5758
ZCOS_pp	158.3131	89.6772	84.6842
ZCAL_ph	150.7909	89.7722	83.4439
ZCALr_ph	150.2952	90.0714	83.7078
ZCOS_ph	150.7885	89.7717	83.4440

It can be seen that the integrated weighting techniques based on person as well as household auxiliary variables perform quite well with the lowest standard errors for Unempl and Sight (and among the lowest for Years5).

Figure 1 shows the summarized percentage relative bias of the estimators, and illustrates the value of using calibration and integration weighting when unequal design weights exist.

- There is a sharp decrease in the percentage relative bias of the estimators UNEMPL and YEARS5 when applying calibration or integrated weighting procedures instead of only the unequal design weights, HWDISTR.
- The integrated weighting techniques based on 'pp' auxiliary information produce slightly better results than the calibration technique, with a further slight improvement when the household auxiliary variable is added ('ph').
- The exception is SIGHT, where the percentage relative bias increases under the integrated techniques, but never rises higher than 5%.

The reason for the strange behaviour of SIGHT can be ascribed to the fact that the correlation between the variable and the auxiliary variables is weak as already indicated in Table 1.

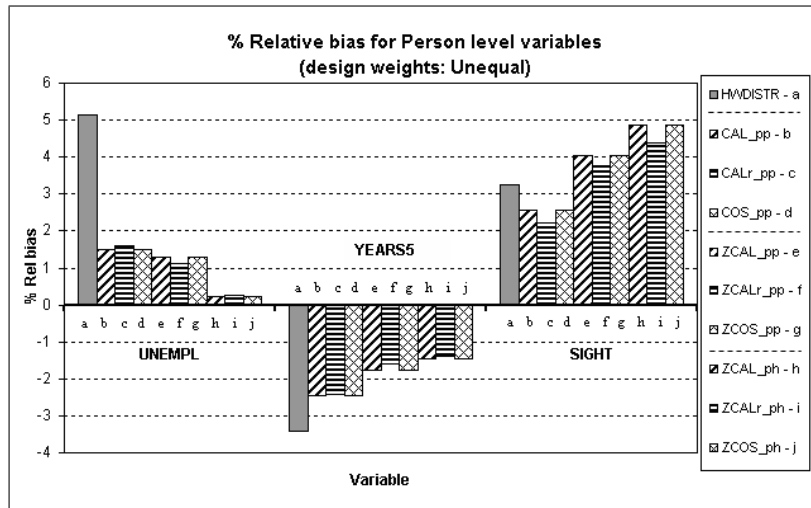


Figure 1. The percentage relative bias of the estimators for the person level variables

5.2 Simulation results of estimators for household level study variables

The effect of using only 'pp' or 'ph' auxiliary information for integrated weighting estimators with unequal design weights for household study variables is evaluated for the person level study variables in Table 6.

It can be concluded that:

- Although most of the biases are positive on average, the minimum bias is negative for the integrated weighting techniques based on 'ph' auxiliary information. Negative bias on average is found for these techniques for COOK.
- By adding the household auxiliary variable to the model, the bias drops substantially.
- The distribution of the bias of the estimators does not differ significantly from the normal distribution.
- The bias of all the estimators differs significantly from zero.

Table 6. Simulation measures obtained for household level variables:
Descriptive statistics of BIAS = (estimated total – population total)

Method	DWEIGR				COOK				WATER			
	Average (AvBias)	max	min	Prob: normal	Average Av(Bias)	max	min	Prob: normal	Average (AvBias)	max	min	Prob: normal
HWDISTR	8134	11805	4803	0.7540	6259	8797	3176	0.4673	4370	6447	1281	0.1514
ZCAL_pp	4603	8248	1483	0.3906	1447	3952	-2001	0.2127	1486	3896	-1866	0.4299
ZCALr_pp	3862	7685	696	0.3293	1257	3770	-2210	0.1844	1302	3623	-2030	0.3773
ZCOS_pp	4600	8245	1481	0.3907	1447	3952	-2001	0.2128	1487	3895	-1866	0.4305
ZCAL_ph	628	3592	1878	0.7085	-219	2438	-2529	0.4656	296	2457	-2508	0.8637
ZCALr_ph	635	3644	1867	0.7029	-207	2517	-2613	0.5406	259	2326	-2521	0.6212
ZCOS_ph	627	3591	1879	0.7118	-219	2437	-2529	0.4661	297	2457	-2507	0.8632

An overall comparison of the integrated weighting estimators for the three household level study variables is shown in Table 7 and Figure 2.

Table 7. Standard errors of simulation study estimators for the totals of household study variables

Method	Std. Error		
	Dwelgrp	Cook	Water
HWDISTR	87.75	71.22	60.17
ZCAL_pp	93.0810	82.5226	69.1035
ZCALr_pp	94.5074	83.2337	69.6693
ZCOS_pp	93.0691	82.5386	69.1080
ZCAL_ph	71.4428	65.8819	60.4737
ZCALr_ph	71.4275	67.2613	61.2771
ZCOS_ph	71.4351	65.8922	60.4761

There is no doubt that the integrated weighting techniques based on person and household auxiliary variables simultaneously yield the lowest standard errors for all three household variables and can therefore be seen as the most stable technique.

As in Table 6, it is clear in Figure 2 that by adding a household auxiliary variable to the model causes the bias of integrated weighting estimators to drop substantially.

Thus it appears integrated weighting improves estimation:

- In view of the standard errors, it may be concluded that the integrated weighting estimators based on 'ph' auxiliary variables generally deliver the most stable estimators.
- A sharp decrease in bias is obtained for all three household study variables when applying 'pp' auxiliary variables under integrated weighting, with a further remarkable decrease when adding the household auxiliary variable.
- Although one of the variables underestimates the population total, the percentage relative bias is negligible when utilizing both person and household auxiliary variables.

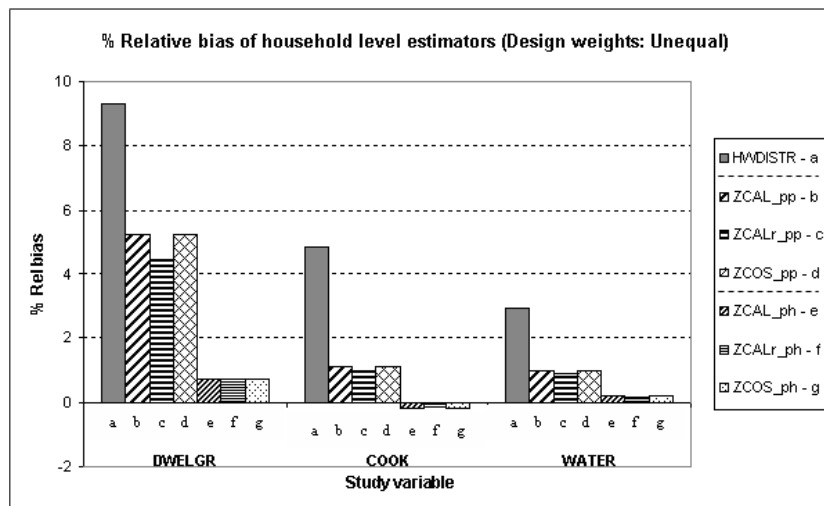


Figure 2. The percentage relative bias of the estimators for the household level variables

To summarize, for both person and household level study variables (except for SIGHT), estimation is improved by applying integrated weighting, especially for the cases where person as well as household auxiliary variables are utilized. The integrated weighting estimates based on person and household auxiliary variables result in the most attractive estimates for all the household study variables under consideration.

6. Conclusions

In practice, researchers are often interested in estimates at different levels. For example, in a household survey the interest could be on person as well as household characteristics, while in business samples the interest may be to obtain information at overall business level, at division level, as well as at employee level. The use of different sets of weights for each level of

information is general practice, but is tedious and may cause a disaster for the quasi- or non-statistical user, who may select the wrong set of weights. Therefore, the main objectives of this study were to evaluate the extension of existing techniques to reconcile person and household weights to a single set of weights. Hence the emphasis was on investigating possible modifications of the integrated weighting technique, based on multivariate auxiliary information for person and household levels simultaneously, and the implication as to bias. To this end, three person level variables were evaluated together with the auxiliary variables, as to bias and variance, for three person level and three household level study variables. The auxiliary variables were either three person level variables, or three person level and one household level variable.

Regarding the different techniques used in this study to estimate the population totals for person level as well as household level study variables, the following overall conclusions can be drawn from the simulation study:

- For most study variables, the standard errors of the integrated weighting techniques, based on person as well as household auxiliary variables were generally the smallest and therefore delivered the more stable estimators.
- Except for SIGHT, the average bias of the integrated weighting estimators decreased substantially from the 'pp' cases, to the cases where person as well as household auxiliary variables were applied. The performance of SIGHT can be attributed to the weak correlation with all the auxiliary variables.
- The average bias of all the estimators differed significantly from zero (Neethling, 2004).

- Regarding the percentage relative bias measures, most of the study variables under consideration yielded a sharp decrease when applying calibration or integrated weighting procedures as opposed to using only the design weight. Furthermore, the integrated weighting technique based on ‘pp’ auxiliary information delivered better results than the calibration techniques, with a further decrease in the percentage relative bias when ‘ph’ auxiliary variables were used. Although the latter case underestimated the population total for some variables, the percentage relative bias was negligible.

In conclusion, it is clear from this study that integration weighting techniques with different levels of auxiliary variables for surveys with different levels of study variables are useful in improving estimation in the sense of reducing bias and providing a single set of weights for the data. Although the difference between the cosmetic estimators and the CAL and ZCAL estimators (with respect to biases and standard errors) was not large in this exploratory study, this is probably due to the fact that only a single stage sampling procedure was considered. An evaluation of the estimators under multistage sampling has not been addressed in the current paper, which explores the advantages of using a single set of weights based on both person and household variables, above the other methods.

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